Internal Assessment Task -II - Study of Ethics in AI and Explainable AI

AI in Healthcare Triage & Clinical Risk Prediction

by

Roll No:16010123324 **Shreya Menon**
Roll No:16010123325 **Shreyans Tatiya**
Roll No:16010123331 **Siddhant Raut**

For the subject

Artificial Intelligence

Department of Computer Engineering
Academic Year 2025-26

AI in Healthcare Triage & Clinical Risk Prediction- Introduction

Artificial intelligence (AI) systems are used in the field of health care to improve clinical triage, risk prediction, and resource allocation. Clinical risk prediction models utilize machine learning algorithms on electronic health records (EHRs) to estimate the probability of certain health issues or patient harm. These predictions are often used to calculate patient risk scores that inform patient prioritization for health care management programs or specialist referrals.

Triage systems in health care are considered to be high-stakes sociotechnical systems. Unlike other AI systems, such as recommendation systems or consumer analytics tools, risk prediction in health care has significant consequences on patient health outcomes and resource allocation. Any inaccuracies or biases in these systems could lead to delayed health care or unfair resource allocation to patients.

The present report presents a critical evaluation of AI in health care triage based on three different dimensions: ethical fairness in health care triage (Obermeyer et al., 2019), explainability and clinical usability in health care triage (Tonekaboni et al., 2019), and responsibility and accountability in medical AI (London, 2019). By incorporating technical evaluation with ethical theory and governance aspects, this report evaluates if the current AI triage systems in health care are responsible.

AI in Healthcare Triage & Clinical Risk Prediction- Case Background

Technical System Description

Healthcare triage systems typically involve supervised learning models trained on historical patient data. The workflow includes:

1. **Data Collection**

   - Demographics
   - Diagnoses (ICD codes)
   - Laboratory results
   - Hospital utilization
   - Prior healthcare spending

2. **Feature Engineering**

   - Chronic condition counts
   - Comorbidity indices
   - Utilization frequency
   - Temporal patterns

3. **Model Training**

   - Objective: minimize prediction error between actual and predicted target variable.
   - In the Obermeyer case, the target variable was future healthcare cost, not illness severity.

4. **Risk Score Output**

   - Continuous risk probability or percentile ranking.
   - Patients above threshold enrolled in care programs.

The Obermeyer Case

In a case study published by Obermeyer et al. in 2019, the authors analyzed a commercial algorithm used in healthcare settings to identify patients who require care management because of their high level of illness. The algorithm used in the case study was optimized to predict healthcare costs, based on the assumption that higher costs indicate a greater level of medical need. However, because of differences in healthcare access, Black patients in the United States have, on average, spent less on healthcare costs than White patients who were equally ill, leading to discrimination in the algorithm's output, even though it was based on proxy data and did not involve any racial discrimination in itself. [2]

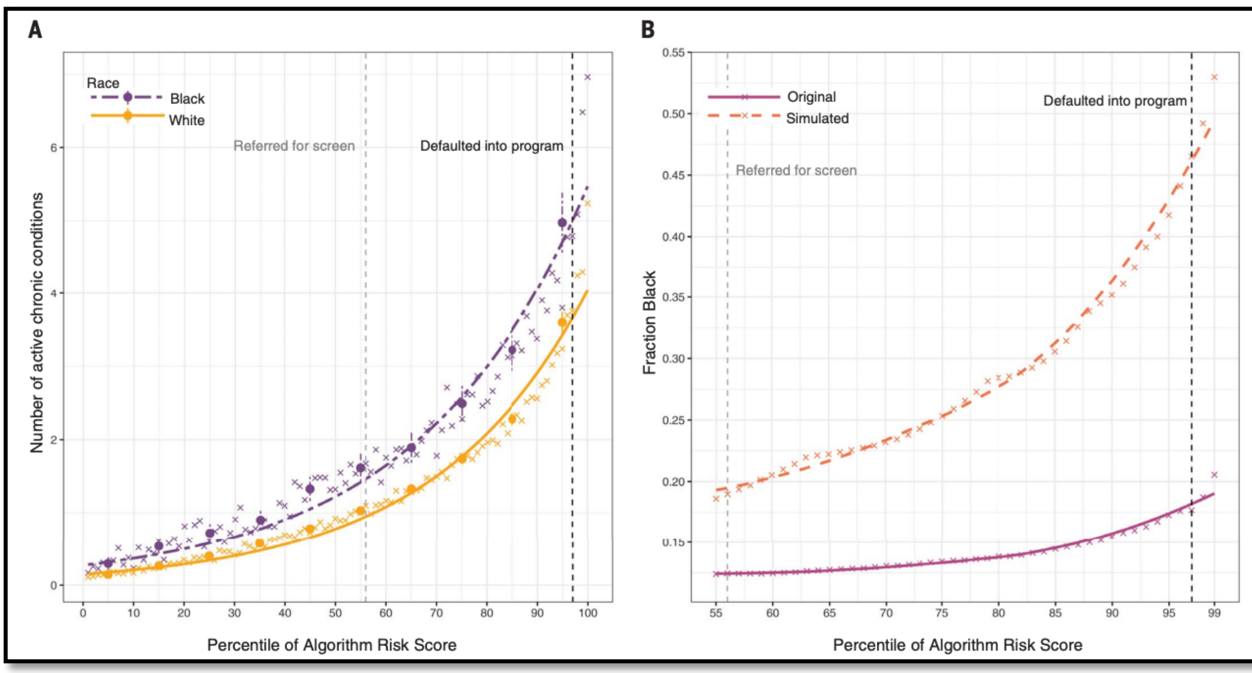

Figure 1. Racial disparity in algorithmic risk scoring
Panel A shows that at equivalent algorithm risk percentiles, Black patients have a higher number of active chronic conditions than White patients. Panel B illustrates how correcting the proxy variable reduces disparity. [2]

AI in Healthcare Triage & Clinical Risk Prediction- Literature Review

In the paper by Obermeyer et al. (2019), the authors provide empirical evidence of racial bias within a widely used healthcare risk prediction algorithm. Through the analysis of claims data, the authors show that the use of healthcare expenditure as a measure of healthcare needs systematically results in an underestimate of the severity of illness among black patients. This paper can be seen as providing a methodological approach to the problem of algorithmic fairness, as well as a specific case that highlights the dangers of misaligned modeling objectives for healthcare triage algorithms.

Tonekaboni et al. (2019) focus on the issue of explainability for clinicians who use machine learning algorithms. Through interviews and analysis, the authors highlight the fact that many of the common techniques for providing explanations, such as feature importance, are not aligned with the cognitive processes of clinicians. Instead, clinicians require explanations that are contextual, case-based, and causally related to the decision-making process. This paper highlights the fact that the issue of interpretability must be sociotechnically constructed to account for the ways in which machine learning algorithms are actually being used.

London (2019) provides a normative analysis of the ethical implications of black box medical decision algorithms. He analyzes the relationship between the predictive accuracy and the level of interpretability of the decision algorithms. He highlights the fact that the black box nature of medical decision algorithms undermines the principles of professional accountability and the requirement for informed consent. He places the issue of interpretability within the broader context of medical ethics, emphasizing the fact that the healthcare practitioner remains morally responsible for the outcome, even when using machine learning algorithms. He frames the issue of interpretability as a requirement for the effective ethical governance of healthcare.

These three papers highlight different dimensions of the use of AI algorithms for healthcare triage. Obermeyer et al. highlight the issue of fairness, Tonekaboni highlight the issue of interpretability, and London highlight the issue of responsibility.

AI in Healthcare Triage & Clinical Risk Prediction - Ethical Analysis

Ethical Framework Analysis: Fairness, Accountability, Transparency, Privacy & Consent, Safety, Explainability, Responsibility

Fairness (Distributive Justice)

Claim: The cost-based triage algorithm violated distributive fairness.

Evidence: Obermeyer et al. demonstrated systematic underestimation of illness burden among Black patients. [2]

Analysis: Distributive justice calls for the equitable distribution of resources among individuals who are comparable. However, the model was not calibrated to meet the needs of the people, only to minimize the cost. Therefore, the model was assuming that the healthcare expenditure was morally neutral. However, the expenditure patterns reveal the structural inequity that exists in the healthcare system. Therefore, the algorithm perpetuated injustice, claiming to be predictive.

Conclusion: Fairness requires alignment between predictive objective and ethical allocation goal.

Accountability

Claim: AI-mediated triage creates responsibility diffusion.

Evidence: London argues that opaque systems obscure the reasoning behind decisions, weakening accountability. [1]

Analysis: In medical ethics, healthcare professionals are answerable for patient outcomes. However, in decision-making processes guided by unexplained algorithmic results, healthcare professionals tend to rely on statistical authority. There is model development, implementation in hospitals, and usage by healthcare professionals.

Conclusion: Governance frameworks must define accountability chains prior to deployment.

Transparency

Tonekaboni et al. demonstrate that clinicians require explanations compatible with diagnostic reasoning. Without transparency, clinicians cannot critically evaluate risk scores or detect anomalies. [3]

Transparency is necessary for:

- Trust calibration
- Bias detection
- Professional justification

Privacy & Consent

The AI triage systems rely on vast amounts of electronic health records, which include diagnostic codes, lab tests, medication, and healthcare utilization. In population health management, these normal health notes and information are used for predictive modeling, often without patients' knowledge that risk scores are being used to determine the course of treatment. For example, in the Obermeyer case, claims information was used to develop risk stratification models that determine the course of treatment. Patients are likely unaware that risk scores, which are calculated using AI and algorithms, are used to determine enrollment in a care management program. This is a second use of health information and informed consent, especially if the predictive models are used to determine access to more effective treatment. These AI triage systems use information from several sources, which increases the risk of function creep, especially if the information is used for predictive modeling. From a moral and ethical perspective, transparency is key to informing patients about how information is used and where AI is used to determine treatment..

Safety

In triage systems:

- False negatives may delay intervention.
- Biased thresholds may disproportionately exclude vulnerable groups.
- Population-scale harm emerges when systemic bias persists.

Responsibility

London argues that AI systems are not morally responsible. Clinicians are responsible for the decision-making process, and hence, AI must not replace the decision-making process. [1]

AI in Healthcare Triage & Clinical Risk Prediction - Explainability Analysis

Explainability Study (XAI Focus)

Black-Box Behavior

As shown in Figure 2, black box models make predictions without revealing their underlying reasoning processes. In the case of the Obermeyer case, it is argued that the black box nature of the model did not result from any intractability of the underlying mathematics, but from a lack of critical inquiry into the optimization objective of the model. The model was trained to predict costs in the future, and this cost was used as a proxy for medical needs. While it is conceivable that information about the features of the model might have been available, the ethical implications of using cost as a proxy for illness severity were never queried.

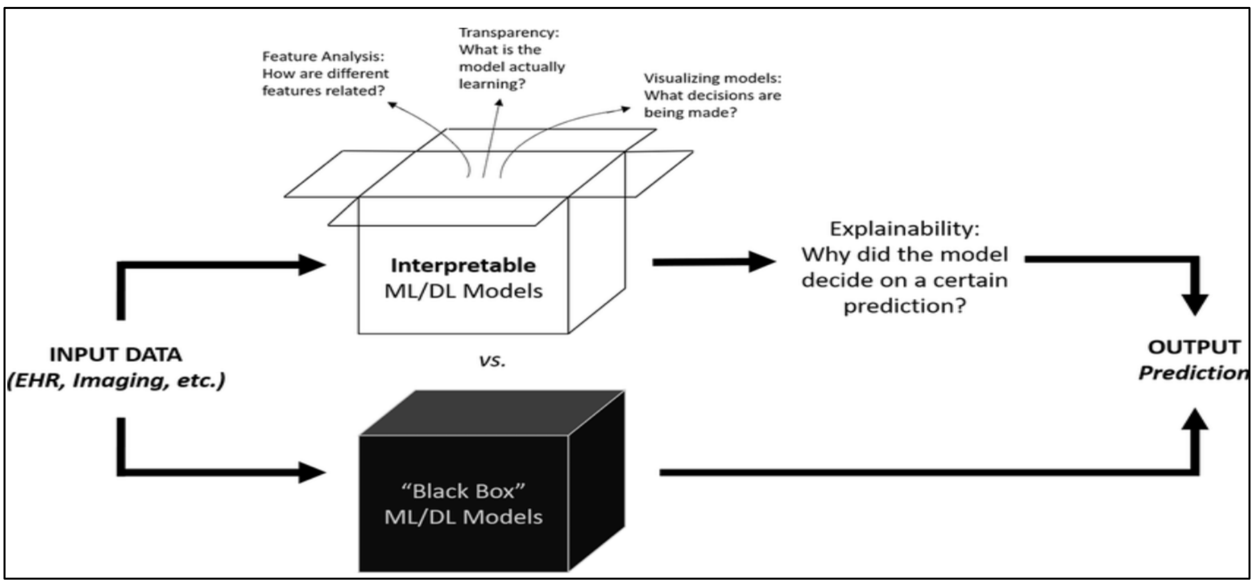

***Figure 2. Visually reinforces the distinction**. While interpretable models allow examination of feature relationships and decision pathways, black-box systems obscure both technical reasoning and normative assumptions. In healthcare, where decisions influence access to care, interpretability is closely tied to accountability. [4]*

Existing Explanation Methods

Common explainability tools include:

- **SHAP** – assigns contribution values to features
- **LIME** – approximates local decision boundaries
- **Partial dependence plots** – show average feature effects
- **Rule-based surrogate models** – approximate model logic

These tools are helpful for increasing statistical transparency but often fail to provide a causally or clinically relevant explanation. As Tonekaboni et al. note, clinicians need contextual reasoning, not abstract feature contributions. Knowing that healthcare spending contributed to the prediction does not tell us if that is ethically good. [3]

In the Obermeyer study, closer examination of SHAP values could have revealed the dominance of prior healthcare spending in shaping predicted risk scores. More importantly, aggregated SHAP analysis across demographic subgroups might have shown that spending-related features reduced predicted risk disproportionately for Black patients compared to White patients with similar illness levels.

Obermeyer et al. demonstrated that replacing healthcare cost with a direct measure of health status reduced racial disparity by nearly 50%, leading to a 50% increase in Black patients identified for care management. This quantifies the impact of target misalignment. However, SHAP values alone would only have revealed the problem; they would not have corrected the flawed objective function. This highlights that explainability must be combined with ethical scrutiny of modeling goals.

How Lack of Explanation Caused Harm

For instance, in the Obermeyer study, the performance of subgroups was not subject to any kind of pre-deployment audit, and if the demographic calibration and feature sensitivity analyses had been conducted, the bias arising due to the use of cost proxies might have been identified earlier. The absence of any kind of explainability also allowed the bias arising due to the use of the proxies to persist even if the model was performing accurately at the group level.

Suitable XAI Enhancements

Improvement strategies include:

- Interpretable models such as generalized additive models (GAMs)
- Counterfactual explanations to test sensitivity to proxy variables
- Subgroup fairness dashboards
- Model documentation outlining objectives and proxy assumptions

However, explainability alone cannot correct a misaligned objective. Even a fully interpretable model will produce inequitable outcomes if it optimizes a biased target variable.

AI in Healthcare Triage & Clinical Risk Prediction - Discussion, Takeaway

First, the optimization objectives must align with the ethical healthcare objectives. In the Obermeyer case, for instance, the technical optimization for the prediction of cost was not necessarily aligned with the equitable provision of healthcare. Therefore, the choice of the optimization objectives is not merely technical; rather, it is ethical.

Second, the issue of explainability must not merely focus on the technical aspect. In the Tonekaboni et al. case, for instance, the issue of explanation was not merely technical, as the clinicians needed explanations that would assist them in the diagnostic process. Therefore, the issue of explainability must not merely focus on the technical aspect, as the clinicians must be able to interpret the results at the individual level and at the level of the population.

Third, the issue of responsibility must not merely focus on the technical aspect. London, for instance, emphasizes that the medical professionals must remain morally responsible, even when the decision-making process is mediated through the use of AI. Therefore, the issue of responsibility must not merely focus on the technical aspect, as the medical professionals must remain morally responsible.

It can therefore be concluded that the healthcare AI must not merely focus on the technical aspect, as the healthcare AI must be contextual, and the issue of responsibility must not merely focus on the technical aspect. In fact, the healthcare AI must be viewed as a sociotechnical artifact that is shaped by the historical data, the institutional practices, and the systemic inequalities.

AI in Healthcare Triage & Clinical Risk Prediction - Controls and Solutions

Regulatory Frameworks

- Mandatory bias audits for high-risk clinical AI.
- Regulatory classification as safety-critical systems.
- Requirement of demographic performance reporting.

Technical Safeguards

- Replace cost proxies with illness severity metrics.
- Continuous monitoring for performance disparities.
- Model retraining under drift conditions.

Transparency Requirements

- Model documentation (model cards).
- Disclosure of training objectives and feature proxies.
- Subgroup fairness dashboards.

Accountability Mechanisms

- Clear responsibility delineation between developers and clinicians.
- Human-in-the-loop review for triage decisions.
- Institutional AI oversight boards.

Recommendations directly address risks identified in fairness, accountability, and transparency analyses.

AI in Healthcare Triage & Clinical Risk Prediction – Conclusion

The analysis of artificial intelligence in healthcare triage and clinical risk prediction has demonstrated that algorithmic systems cannot and should not be assessed on the grounds of predictive accuracy. For example, in the case of the Obermeyer case study, the optimization process was geared towards cost reduction rather than medical need. This has been demonstrated to lead to systemic racial disparities in the allocation of care. This has been attributed to objective misalignment and proxy variables. This has demonstrated that model targets and loss functions are ethical decisions and not technical decisions.

The study conducted by Tonekaboni et al. has demonstrated that explainability in healthcare should not only include statistical feature attribution. Rather, explainability should include contextual and workflow-integrated explanations to support diagnostic reasoning and error detection. This has demonstrated that explainability not only acts as a transparency tool but also acts as a tool for accountability and trust calibration.

The study conducted by London has demonstrated that accountability in medical AI cannot and should not be left to algorithms. Rather, clinicians and institutions have moral and professional accountability for patient outcomes.

This has been demonstrated to lead to governance gaps and diffuse liability in the absence of well-defined accountability.

References

1. London, A. J. (2019). Artificial intelligence and black-box medical decisions: Accuracy versus explainability. *Hastings Center Report, 49*(1), 15–21. https://doi.org/10.1002/hast.973

2. Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage the health of populations. *Science, 366*(6464), 447–453. https://doi.org/10.1126/science.aax2342

3. Tonekaboni, S., Joshi, S., McCradden, M. D., & Goldenberg, A. (2019). What clinicians want: Contextualizing explainable machine learning for clinical end use. *Proceedings of Machine Learning Research, 106*, 359–380. https://proceedings.mlr.press/v106/tonekaboni19a.html

4. Ethical challenges of artificial intelligence in health care: A narrative review – Scientific figure on ResearchGate. (2026). *Black-box AI models versus interpretable and explainable AI models* [Figure]. ResearchGate. Retrieved March 3, 2026, from https://www.researchgate.net/figure/Black-box-AI-models-versus-interpretable-and-explainable-AI-models-DL-deep-learning_fig2_357757904

Conclusion and take away from this exercise

A primary inference of the above analysis is that the artificial intelligence systems deployed in the domain of health care are sociotechnical artifacts that are influenced by historical data, existing institutional practices, and structural inequalities. For the ethical use of AI, alignment of objectives with health care values, auditing of demographic data beyond the use of accuracy, and human oversight of AI systems are critical requirements. Although the use of explainability tools promotes the use of transparency, it does not eliminate the issue of the alignment of objectives and biases inherent in the data sets.

Hence, the use of AI for health care requires a comprehensive approach that incorporates technical, ethical, and institutional aspects of health care. In the specific case of AI-based triage systems, the use of such tools necessitates the consideration of the values of equity, accountability, and dignity, apart from the use of other metrics for the evaluation of AI-based health care tools.